

Solar Si appendix

Sun · Si · downloadable appendix

Solar silicon: 7.489 ± 0.074

Asplund, Amarsi & Grevesse 2021: 7.51 ± 0.03 · this measurement -0.021 dex

Kitt Peak (1984, molecfit) · VIS · Reference Grade · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · $n = 7$.
The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

7.489 ± 0.074

± 0.020 stat · ± 0.071 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

7.578 ± 0.127

± 0.035 stat · ± 0.122 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

7.549 ± 0.086

± 0.030 stat · ± 0.081 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.325 ± 0.090

± 0.029 stat · ± 0.085 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · CRIRES+ H · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	42 products -- Si I · Asplund et al. 2021 lines; Si I · Deshmukh et al. 2022 lines; Si I · Elgueta et al. 2026 lines; Si II · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	27 products -- Si I · Codex-graded lines; Si II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ± 0.142 stat ± 0.098 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.511 ± 0.024 stat ± 0.100 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.514 ± 0.047 stat ± 0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.487 ± 0.048 stat ± 0.072 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.514 ± 0.047 stat ± 0.065 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.487 ± 0.048 stat ± 0.072 syst

VIS — IAG FTS A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.519 ± 0.094 stat ± 0.103 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.489 ± 0.105 stat ± 0.102 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.511 ± 0.015 stat ± 0.070 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.484 ± 0.016 stat ± 0.076 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.511 ± 0.015 stat ± 0.070 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.484 ± 0.016 stat ± 0.076 syst

VIS — Kitt Peak (Kurucz 2005) A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.528 ± 0.144 stat ± 0.102 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.491 ± 0.311 stat ± 0.102 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.513 ± 0.020 stat ± 0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.486 ± 0.021 stat ± 0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.513 ± 0.020 stat ± 0.065 syst

VIS — Kitt Peak (Kurucz 2005) A(Si) dex; axis 7.15–7.83

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 7 lines



7.486
±0.021 stat ±0.071 syst

VIS — Kitt Peak (1984, molecfit) A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 1 line



7.518
±0.110 stat ±0.110 syst

Si II · Codex-graded lines · 1D LTE
Codex Grade · 3 lines



7.488
±0.197 stat ±0.106 syst

Si I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 7 lines



7.516
±0.019 stat ±0.064 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)
Reference Grade · 7 lines



7.489
±0.020 stat ±0.071 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 7 lines



7.516
±0.019 stat ±0.064 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 7 lines



7.489
±0.020 stat ±0.071 syst

VIS axis: 7.15 · 7.32 · 7.49 · 7.66 · 7.83 A(Si) dex

red-optical — IAG FTS A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 2 lines



7.591
±0.030 stat ±0.121 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)
Reference Grade · 2 lines



7.573
±0.028 stat ±0.125 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 2 lines



7.573
±0.028 stat ±0.125 syst

red-optical — Kitt Peak (Kurucz 2005) A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 2 lines



7.581
±0.035 stat ±0.120 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)
Reference Grade · 2 lines



7.563
±0.033 stat ±0.123 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 2 lines



7.563
±0.033 stat ±0.123 syst

red-optical — Kitt Peak (1984, molecfit) A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 2 lines



7.596
±0.037 stat ±0.118 syst

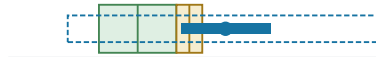
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)
Reference Grade · 2 lines



7.578
±0.035 stat ±0.122 syst

red-optical — Kitt Peak (1984, molecfit) A(Si) dex; axis 7.41–7.74

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 2 lines



7.578
±0.035 stat ±0.122 syst

red-optical axis: 7.41 · 7.49 · 7.58 · 7.66 · 7.74 A(Si) dex

NIR — CRIRES+ J A(Si) dex; axis 7.26–7.77

Si I · Deshmukh et al. 2022 lines · 1D LTE
Reference Grade · 2 lines



7.513
±0.005 stat ±0.086 syst

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi)
Reference Grade · 2 lines



7.487
±0.009 stat ±0.091 syst

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 9 lines



7.518
±0.031 stat ±0.084 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 9 lines



7.496
±0.030 stat ±0.089 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 27 lines



7.525
±0.045 stat ±0.171 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 9 lines



7.496
±0.030 stat ±0.089 syst

NIR — CRIRES+ Y A(Si) dex; axis 7.26–7.77

Si I · Deshmukh et al. 2022 lines · 1D LTE
Reference Grade · 1 line



7.468
±0.018 stat ±0.135 syst

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi)
Reference Grade · 1 line



7.428
±0.018 stat ±0.139 syst

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 2 lines



7.470
±0.002 stat ±0.096 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 2 lines



7.442
±0.015 stat ±0.101 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 4 lines



7.584
±0.069 stat ±0.102 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 2 lines



7.442
±0.015 stat ±0.101 syst

NIR — CRIRES+ Y (full arm) A(Si) dex; axis 7.26–7.77

Si I · Deshmukh et al. 2022 lines · 1D LTE
Reference Grade · 1 line



7.468
±0.018 stat ±0.135 syst

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi)
Reference Grade · 1 line



7.428
±0.018 stat ±0.139 syst

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 5 lines



7.471
±0.020 stat ±0.065 syst

NIR — CRIRES+ Y (full arm) A(Si) dex; axis 7.26–7.77

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 5 lines



7.458
±0.021 stat ±0.072 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 8 lines



7.551
±0.056 stat ±0.073 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 5 lines



7.458
±0.021 stat ±0.072 syst

NIR — IAG FTS A(Si) dex; axis 7.26–7.77

Si I · Deshmukh et al. 2022 lines · 1D LTE
Reference Grade · 1 line



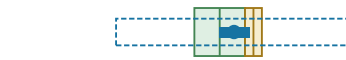
7.491
±0.092 stat ±0.136 syst

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi)
Reference Grade · 1 line



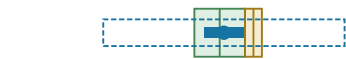
7.451
±0.092 stat ±0.139 syst

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 4 lines



7.527
±0.018 stat ±0.140 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 4 lines



7.515
±0.023 stat ±0.143 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 18 lines



7.538
±0.070 stat ±0.105 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 4 lines



7.515
±0.023 stat ±0.143 syst

NIR — Kitt Peak (Kurucz 2005) A(Si) dex; axis 7.26–7.77

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 1 line



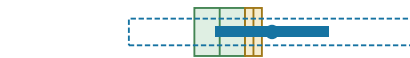
7.558
±0.098 stat ±0.145 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 1 line



7.546
±0.098 stat ±0.148 syst

Si I · Codex-graded lines · 1D LTE
Codex Grade · 14 lines



7.572
±0.067 stat ±0.170 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 1 line



7.546
±0.098 stat ±0.148 syst

NIR — Kitt Peak (1984, molecfit) A(Si) dex; axis 7.26–7.77

Si I · Deshmukh et al. 2022 lines · 1D LTE
Reference Grade · 3 lines



7.581
±0.030 stat ±0.069 syst

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi)
Reference Grade · 3 lines



7.552
±0.035 stat ±0.075 syst

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 15 lines



7.556
±0.030 stat ±0.075 syst

NIR — Kitt Peak (1984, molecfit) A(Si) dex; axis 7.26–7.77

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 15 lines



7.549
±0.030 stat ±0.081 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 15 lines



7.549
±0.030 stat ±0.081 syst

NIR axis: 7.26 · 7.39 · 7.52 · 7.64 · 7.77 A(Si) dex

H — CRIRES+ H A(Si) dex; axis 7.21–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE
Reference Grade · 3 lines



7.363
±0.031 stat ±0.080 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi)
Reference Grade · 3 lines



7.325
±0.029 stat ±0.085 syst

Si I · Codex-graded lines · 1D NLTE (Amarsi)
Codex Grade · 3 lines



7.325
±0.029 stat ±0.085 syst

H axis: 7.21 · 7.30 · 7.40 · 7.49 · 7.59 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
VIS	Si I · Codex-graded lines; Si II · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
VIS	Si I · Asplund et al. 2021 lines; Si II · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
red-optical	Si I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-NLTE	2	7.563–7.578 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.03
red-optical	Si I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	7.563–7.578 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.03
NIR	Si I · Codex-graded lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–27	7.442–7.549 (1D-NLTE)	telluric residual 0.05, line strengths (gf) 0.04
NIR	Si I · Deshmukh et al. 2022 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	7.428–7.552 (1D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
NIR	Si I · Elgueta et al. 2026 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–15	7.442–7.549 (1D-NLTE)	telluric residual 0.05, line strengths (gf) 0.04
H	Si I · Codex-graded lines	CRIRES+ H	1D-NLTE	3	7.325 (1D-NLTE)	line strengths (gf) 0.05, fit window 0.05
H	Si I · Elgueta et al. 2026 lines	CRIRES+ H	1D-LTE, 1D-NLTE	3	7.325 (1D-NLTE)	line strengths (gf) 0.05, fit window 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 59 lines.

Abundance outlier: disagrees with the other lines well beyond its error, and no solar analysis in the literature uses it.

Not measurable: the line falls in a data gap or a saturated telluric band.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width $\log W/\lambda$ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Si II	3203.872	saturated ($\log W/\lambda = -4.18$)
Si II	3210.028	saturated ($\log W/\lambda = -3.98$)
Si II	3339.818	saturated ($\log W/\lambda = -4.04$)
Si II	3856.017	saturated ($\log W/\lambda = -3.90$)
Si II	3862.595	saturated ($\log W/\lambda = -4.15$)
Si I	3905.523	saturated ($\log W/\lambda = -3.85$)
Si I	4102.936	saturated ($\log W/\lambda = -4.10$)

Species	Wavelength (Å, air)	Why it is excluded
Si I	5948.541	saturated (log W/λ = -4.79)
Si I	7918.383	saturated (log W/λ = -4.88)
Si I	7932.348	saturated (log W/λ = -4.81)
Si I	7944.001	saturated (log W/λ = -4.75)
Si I	8543.151	saturated (log W/λ = -4.23)
Si I	8648.465	saturated (log W/λ = -4.72)
Si I	8661.177	saturated (log W/λ = -4.27)
Si I	8728.010	saturated (log W/λ = -4.89)
Si I	8742.446	saturated (log W/λ = -4.90)
Si I	8752.007	saturated (log W/λ = -4.79)
Si I	8790.389	saturated (log W/λ = -4.82)
Si I	9021.590	saturated (log W/λ = -4.88)
Si I	9413.500	saturated (log W/λ = -4.56)
Si I	9887.047	abundance outlier
Si I	10313.197	abundance outlier
Si I	10371.263	saturated (log W/λ = -4.73)
Si I	10585.141	saturated (log W/λ = -4.46)
Si I	10603.425	saturated (log W/λ = -4.57)
Si I	10627.648	saturated (log W/λ = -4.85)
Si I	10660.973	saturated (log W/λ = -4.53)
Si I	10689.716	saturated (log W/λ = -4.70)
Si I	10694.251	saturated (log W/λ = -4.66)
Si I	10727.406	saturated (log W/λ = -4.62)
Si I	10749.378	saturated (log W/λ = -4.54)
Si I	10786.849	saturated (log W/λ = -4.59)
Si I	10827.088	saturated (log W/λ = -4.40)
Si I	10843.858	saturated (log W/λ = -4.73)
Si I	10868.790	saturated (log W/λ = -4.49)
Si I	10869.536	saturated (log W/λ = -4.34)
Si I	10885.333	saturated (log W/λ = -4.84)
Si I	10979.308	saturated (log W/λ = -4.62)
Si I	10982.058	saturated (log W/λ = -4.90)
Si I	11017.966	saturated (log W/λ = -4.68)
Si I	11187.588	saturated (log W/λ = -4.89)
Si I	11289.829	saturated (log W/λ = -4.86)
Si I	11611.060	saturated (log W/λ = -4.60)
Si I	11984.198	saturated (log W/λ = -4.46)
Si I	11991.568	saturated (log W/λ = -4.51)
Si I	12031.504	saturated (log W/λ = -4.38)
Si I	12103.534	saturated (log W/λ = -4.59)
Si I	12189.241	abundance outlier
Si I	12270.692	saturated (log W/λ = -4.62)
Si I	15077.332	saturated (log W/λ = -4.67)
Si I	15557.778	saturated (log W/λ = -4.88)
Si I	15888.409	saturated (log W/λ = -4.39)
Si I	15960.063	saturated (log W/λ = -4.53)

Species	Wavelength (Å, air)	Why it is excluded
Si I	16060.009	saturated (log W/λ = -4.64)
Si I	16163.691	saturated (log W/λ = -4.78)
Si I	16380.176	saturated (log W/λ = -4.72)
Si I	16381.534	saturated (log W/λ = -4.54)
Si I	16434.927	not measurable
Si I	17466.879	saturated (log W/λ = -4.45)

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

[Published line sets](#) · [gf uncertainty model](#) · [Methodology](#)

Download the evidence

[Download PDF — Solar Si appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned science snapshot. Source commit c730d5e72e3f8902d1d4d800b46fe89fbcafee69

Feed Si.json v1.165 · updated 2026-10-09T22:31:47Z

Generated 2026-10-09T16:35:54-06:00 (source commit timestamp)

Feed SHA-256 f5230bddfe2fd5b1062d15db46d75291aa384dd706693a30c9ec3062e31f81ec

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

[Science and provenance source notes](#)

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [\[3\]](#).

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [\[6\]](#).

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [8].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [10].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [11].

Product-to-source index

VIS · HARPS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-18313ebf6e74

VIS · HARPS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-a600a1f3ca00

VIS · IAG FTS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7b7e31f61f2b

VIS · IAG FTS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-d538a2159904

VIS · Kitt Peak (Kurucz 2005) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6]
Si-products-0d051b970724

VIS · Kitt Peak (Kurucz 2005) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6]
Si-products-c60cd632806c

VIS · Kitt Peak (1984, molecfit) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6]
Si-products-45b0eb7e0d7e

VIS · Kitt Peak (1984, molecfit) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6]
Si-products-eb4fab3570b0

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-5005aa52ea70

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-c6dd0223e3f3

H · CRIRES+ H · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-8d77b8828ac4

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-9fbe2e5b1658

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10]
Si-products-1ded76ce0fad

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-8c9605b0a639

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-cb294609df83

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-11a4bee70bc1

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-370653c5a51b

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-e59dc7543e5f

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10]
Si-products-88b8c28f5e69

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-1abc6a5b7f7e

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-18dbbb90d589

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-c1e9abaf4912

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-cd83dc78cee8

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6]
Si-products-9ef011bb7736

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10]
Si-products-f7c09fec6eba

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6]
Si-products-5423ebdd0afb

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-09161c70025a

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-faeb4cf3a572

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10]
Si-products-47c3bb6f832b

NIR · IAG FTS · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-e785ba3e08c1

NIR · IAG FTS · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10]
Si-products-04f1e24adbb8

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-81f4ee1a5870

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-896472fe050f

NIR · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-c8b40a48647f

NIR · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-0833fbcc7bb7

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6]
Si-products-3a10ab9b620b

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-188eb4d27452

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-10d9018840e0

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10]
Si-products-ae3ff274c43d

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6]
Si-products-b89b1c1c0cc7

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10]
Si-products-6aea841c5ab0

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6]
Si-products-029038f614bd

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10]
Si-products-22a294d44708

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10]
Si-products-ec194c374afc

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-876b0e3862e2

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10]
Si-products-51b87205ed62

VIS · HARPS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-cd5ad9c55a7a

VIS · HARPS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-4cca248d4fbd

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7f156df3588f

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-2a3bbd767593

VIS · IAG FTS · Si I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#) Si-products-a993b62b6610

VIS · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-21c709670d7f

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-636d77dd1354

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-bd2f9077b05e

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-cb7c576a5951

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-87c7629c14db

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-9b123c2ddb3d

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-89f923ebea1f

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-513e0cea2d6a

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-c67227338c58

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-a4ee421ad3ac

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-86b202a873f0

red-optical · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-0c1a08dd2576

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-e077c640e18d

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#) Si-products-1c8b249eb403

red-optical · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-bdf98214dcc6

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)
Si-products-bc08240c5f04

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#) Si-products-b848e68841d2

red-optical · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[10\]](#)
Si-products-dbec0948cd03

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. *Space Science Reviews* 221 DOI: 10.1007/s11214-025-01146-w. [source record](#)

[3] — Amarsi, A. M.; Asplund, M. (2017). The solar silicon abundance based on 3D non-LTE calculations. *MNRAS* 464, 264-273 DOI: 10.1093/mnras/stw2445. [source record](#)

[8] — Elgueta, S.; Jofre, P.; Aguilera-Gomez, C.; Slumstrup, D.; Rojas Arriagada, A.; Heiter, U.; Casamiquela, L.; Zoccali, M.; Worley, C.; Soubiran, C. (2026). Gaia FGK benchmark stars: Selecting infrared lines for abundance determination. *A&A* 710, A111 DOI: 10.1051/0004-6361/202659148. [source record](#)

[11] — Deshmukh, S. A.; Ludwig, H.-G.; Kucinkas, A.; et al. (2022). The solar photospheric silicon abundance according to CO5BOLD: Investigating line broadening, magnetic fields, and model effects. *A&A* 668, A48 DOI: 10.1051/0004-6361/202142072. [source record](#)

Atomic data and gf authorities

[4] — Garz, T. (1973). Absolute oscillator strengths of Si I lines between 2500 Å and 8000 Å. *A&A* 26, 471-477 [source record](#)

[5] — O'Brian, T. R.; Wickliffe, M. E.; Lawler, J. E.; Whaling, W.; Brault, J. W. (1991). Lifetimes, transition probabilities, and level energies in Fe I. *JOSA B* 8, 1185 DOI: 10.1364/JOSAB.8.001185. [source record](#)

[7] — Kelleher, D. E.; Podobedova, L. I. (2008). Atomic Transition Probabilities of Silicon. A Critical Compilation. *J. Phys. Chem. Ref. Data* 37, 1285-1501 DOI: 10.1063/1.2734566. [source record](#)

[9] — Pehlivan Rhodin, A.; Hartman, H.; Nilsson, H.; Joensson, P. (2024). Accurate and experimentally validated transition data for Si I and Si II. *A&A* 682, A184 DOI: 10.1051/0004-6361/202245686. [source record](#)

Model and grid sources

[6] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. *A&A* 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

[10] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. *A&A* 642, A62 DOI: 10.1051/0004-6361/202038650. [source record](#)